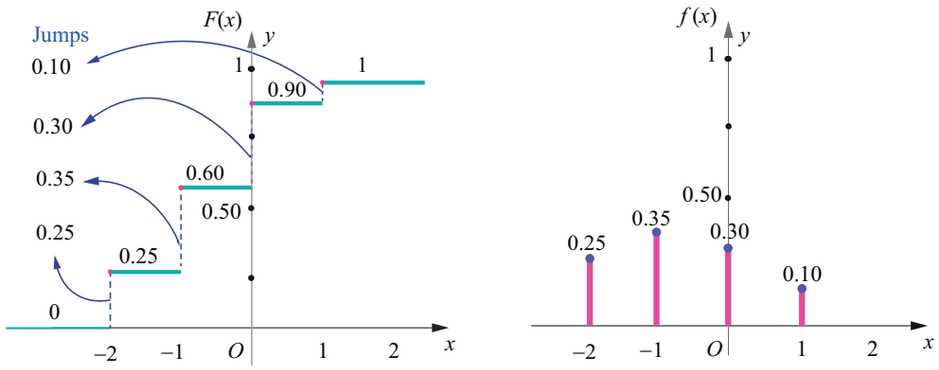



Distribution function $F(x)$ and jumps at each of $x_i$

Probability mass function $f(x)$

Fig. 11.10

(i) $P(X<0)=\sum_{-\infty}^{-1}P(X=x)=P(X=-2)+P(X=-1)=0.25+0.35=0.60$.

(ii) $P(X\geq -1)=\sum_{-1}^{1}P(X=x)=P(X=-1)+P(X=0)+P(X=1)=0.35+0.30+0.10=0.75$

## Example 11.10

A random variable $X$ has the following probability mass function.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(x)$ | $k$ | $2k$ | $6k$ | $5k$ | $6k$ | $10k$ |

Find (i) $P(2<X<6)$ (ii) $P(2\leq X<5)$ (iii) $P(X\leq 4)$ (iv) $P(3<X)$

## Solution

Since the given function is a probability mass function, the total probability is one. That is $\sum_x f(x)=1$.

From the given data $k+2k+6k+5k+6k+10k=1$

$$30k=1 \Rightarrow k=\frac{1}{30}$$

Therefore the probability mass function is

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $f(x)$ | $\frac{1}{30}$ | $\frac{2}{30}$ | $\frac{6}{30}$ | $\frac{5}{30}$ | $\frac{6}{30}$ | $\frac{10}{30}$ |

(i) $P(2<X<6)=f(3)+f(4)+f(5)=\frac{6}{30}+\frac{5}{30}+\frac{6}{30}=\frac{17}{30}$.

(ii) $P(2\leq X<5)=f(2)+f(3)+f(4)=\frac{2}{30}+\frac{6}{30}+\frac{5}{30}=\frac{13}{30}$.

(iii) $P(X\leq 4)=f(1)+f(2)+f(3)+f(4)=\frac{1}{30}+\frac{2}{30}+\frac{6}{30}+\frac{5}{30}=\frac{14}{30}$.

(iv) $P(3<X)=f(4)+f(5)+f(6)=\frac{5}{30}+\frac{6}{30}+\frac{10}{30}=\frac{21}{30}$.



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